Best Practice

IUS Technical Complaints Guidance document

1429

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Applicability

Division: PH

Area(s): Quality

This document is electronically approved (signatures on the last page)

Distribution Sheet for Quality Regulations

Division	PH
Product Lifecycle Phase	Commercial Operation, Manufacturing
Region/ Country*	Global
Product Group**	Sterile drug products (incl. bulk and combination products)
Relevance for PH R&D	N/A

* *Manufactured in or for region / country*
** *Regulation is distributed to each site that handles the product group*

Lead Area:	Quality
Global Process Owner:	Andreas Joos (GBEDL)
Document Author(s):	Mirja Kiiskinen
Process Group	Manage Quality Issue
Document level	Best Practice to GSOP 115

1 Scope

This Best Practice supports the standardized description of defects and classification according to GSOP 115 “Manage Product Technical Complaint and Suspected Falsification and Counterfeit Incident” and provides guidance to enter IUS family product complaints into the complaint database.

The IUS (Intrauterine System) products are:

- Mirena
- Jaydess/Skyla
- Kyleena

This document provides guidance on the following:

- IUS family Product Description
- Common IUS Complaints
- Definitions of Common IUS Complaints
- Complaint database Object Types
- Complaint database Error Codes
- Complaint database Error Sub-Codes
- Whether or not a complaint sample is requested
- Initial risk classification
- LODE case handling

2 Abbreviations

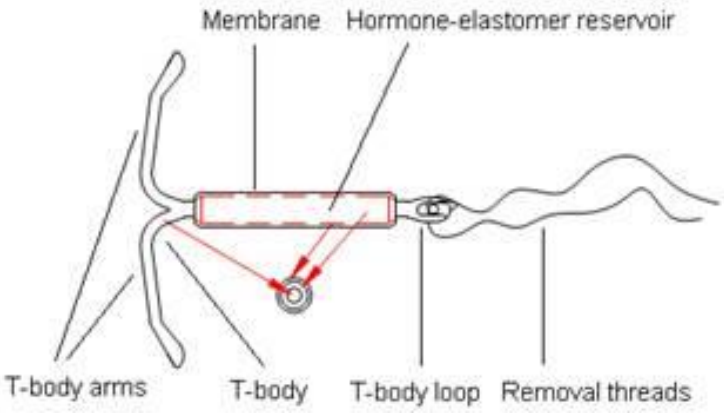
AE	Adverse Event
CQH	Country Quality Head
CSI	Customer Satisfaction Issue
HCP	Health Care Provider or Professional
IUS	Intrauterine System
LODE	Lack of Drug Effect
PTC	Product Technical Complaint
RQU	Responsible Quality Unit
UE	Use Error
UI	Usability Issue

Additionally, definitions of various kinds of complaints and used abbreviations are available in global complaint handling GSOP 115 “Manage Product Technical Complaint and Suspected Falsification and Counterfeit Incident”.

3 IUS Product Description

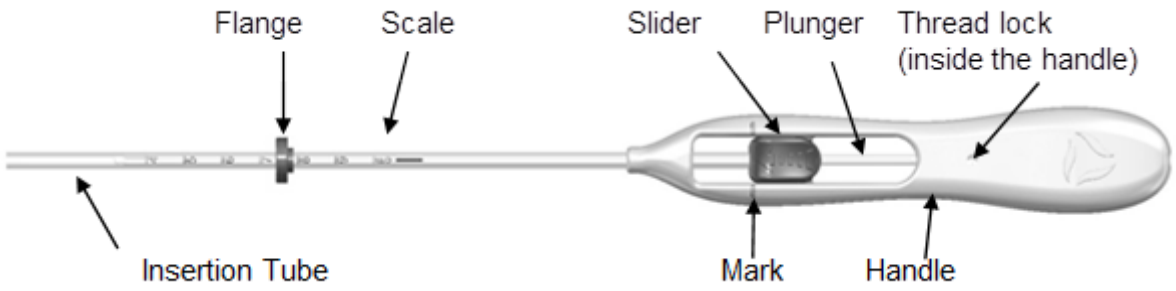
IUS – The IUS consist of a soft, flexible T-body, silver ring (only Kyleena and Jaydess/Skyla) and hormone cylinder which releases small amounts of hormone locally into the female uterus (see pic. 1 IUS components).

Figure 1: IUS components



EvoInserter – All IUS products are provided with an inserter that is used for insertion of the unit into the uterine cavity. The inserter consists of a symmetrical two-sided handle, pre-bent insertion tube, flange, slider and plunger (see pic 2. Inserter components). The inserter functionality is identical for all IUS products. Once the IUS is inserted, the inserter is discarded. The device is for single use only. Once released, the device cannot be reloaded.

Figure 2: Inserter components



3.1 Product Description

Mirena (levonorgestrel-releasing intrauterine system) consists of a flexible T-shaped polyethylene frame (T-body) with a steroid reservoir (hormone elastomer core) wrapped around the vertical stem. The reservoir consists of a cylinder, made of a mixture of levonorgestrel (52 mg) and silicone (polydimethylsiloxane) membrane. The T-body is 32 mm in both the horizontal and vertical directions. The polyethylene used in the T-body is compounded with barium sulfate, making it radiopaque. A monofilament brown polyethylene removal thread is attached to a loop at the end of the vertical stem of the T-body.

Jaydess / Skyla (levonorgestrel-releasing intrauterine system) consists of a flexible T-shaped polyethylene frame (T-body) with a steroid reservoir (hormone elastomer core) wrapped around the vertical stem. The reservoir consists of a cylinder, made of a mixture of levonorgestrel (13.5 mg) and silicone (polydimethylsiloxane) membrane. The T-body is 28 mm in the horizontal direction and 30 mm in the vertical directions. The polyethylene used in the T-body is compounded with barium sulfate,

making it radiopaque. The vertical stem contains a silver ring. A monofilament brown polyethylene removal thread is attached to a loop at the end of the vertical stem of the T-body.

Kyleena (levonorgestrel-releasing intrauterine system) consists of a flexible T-shaped polyethylene frame (T-body) with a steroid reservoir (hormone elastomer core) wrapped around the vertical stem. The reservoir consists of a cylinder, made of a mixture of levonorgestrel (19.5 mg) and silicone (polydimethylsiloxane) membrane. The T-body is 28 mm in the horizontal direction and 30 mm in the vertical directions. The polyethylene used in the T-body is compounded with barium sulfate, making it radiopaque. The vertical stem contains a silver ring. A monofilament blue polypropylene removal thread is attached to a loop at the end of the vertical stem of the T-body.

Figure 3: Top Mirena, in the middle Jaydess and in the bottom Kyleena. The IUS is inside the inserter and the product is in starting position, slider at the mark line

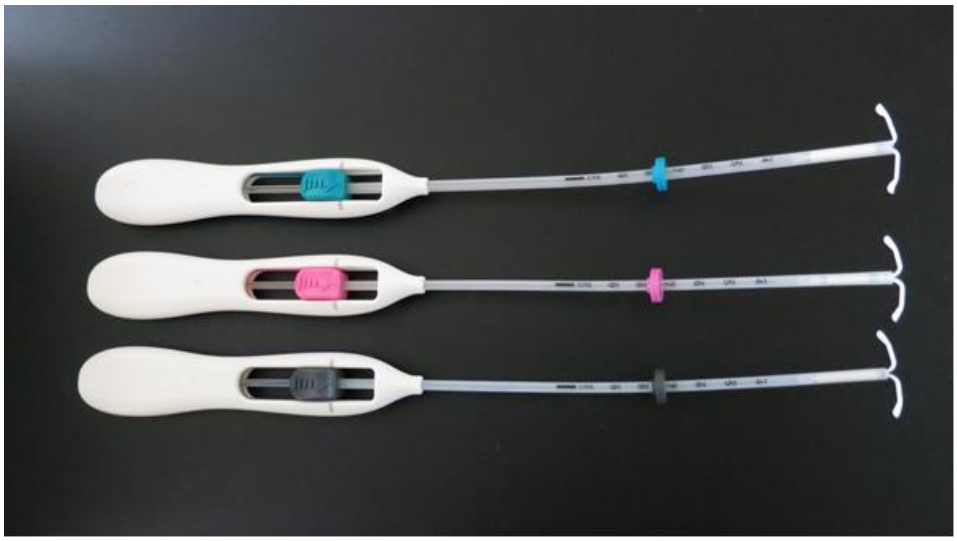
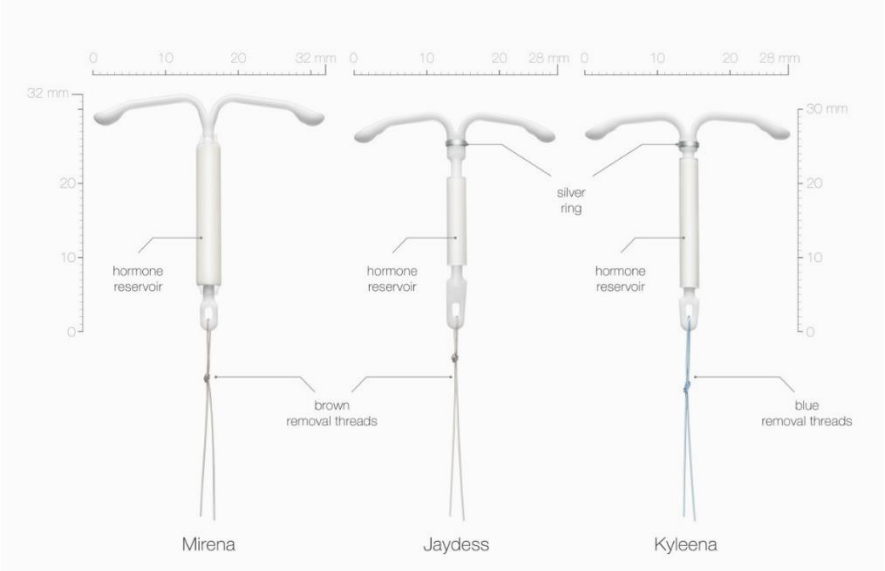
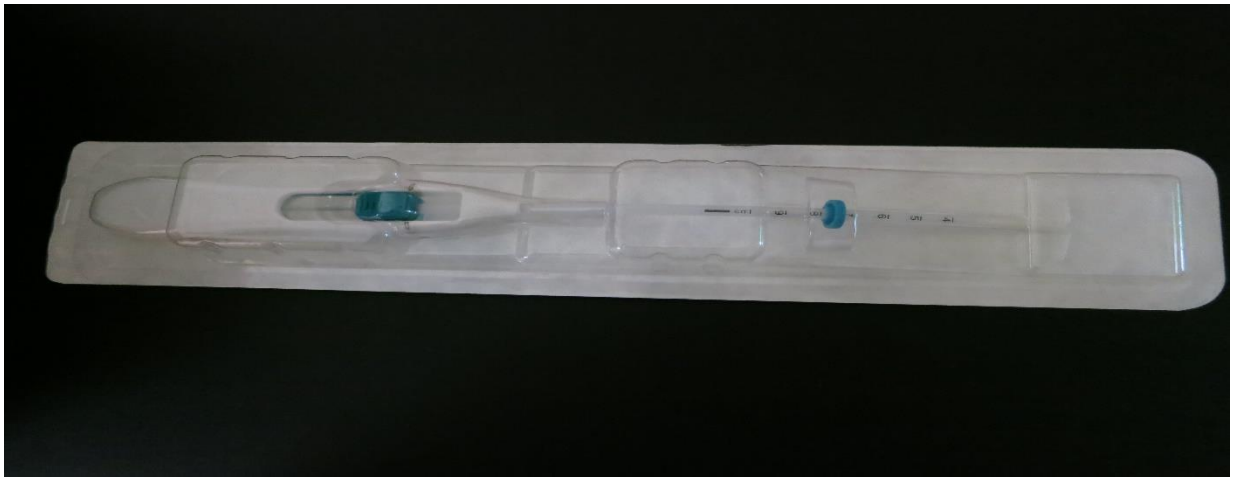


Figure 4: IUSs with threads and their differences



3.2 Insertion Process

Figure 5: Mirena IUS with inserter in primary package



Opening of Package – Since the contents of the package are sterile, care must be taken when opening the package and when preparing to load the unit. The package is opened completely. Use sterile technique and sterile gloves during insertion procedure.

Loading – Push the slider forward in the direction of the arrow to the furthest position to load Mirena into the insertion tube. Holding the slider in the furthest position, set the upper edge of the flange to correspond to the sound measurement of the uterine depth.

Insertion – While holding the slider in the furthest position, advance the inserter through the cervix until the flange is approx. 1.5-2.0 cm from the uterine cervix. While holding the inserter steady, pull the slider to the mark to open the horizontal arms of Mirena. Wait 5 -10 seconds for the horizontal arms to open completely. Advance the inserter gently towards the fundus of the uterus until the flange touches the cervix. Mirena is now in the fundal position.

Release and Withdrawal – Holding the inserter in place, release Mirena by pulling the slider all the way down. While holding the slider all the way down, gently remove the inserter by pulling it out.

Cutting of Threads – Cut the threads to leave about 2-3 cm visible outside of the cervix. Threads should be cut perpendicular using sharp, curved scissors, leaving about 2-3 cm visible outside the cervix. Care should be taken not to apply tension or pull on the threads to prevent displacing the unit.

Removal of IUS – Mirena is removed by pulling on the threads with forceps.

4 IUS - Product Technical Complaints

4.1 Case Entry Instructions for IUS-Related Complaints

1. **Separate Case Entries:**
   - Each IUS-related case must be entered as a separate case entry in the complaint database, regardless of whether they originate from the same Health Care Provider (HCP), the same batch, or involve the same complaint reason.
   - Each device is considered an individual case that requires independent investigation and reporting.
2. **Reporting Packaging Manufacturing Defects:**
   - Complaints related to the same packaging manufacturing defect may be reported together, even if they involve multiple packages and batches.
3. **Classification of Cases**:

   Cases must be processed as Product Technical Complaints (PTCs) if any of the following conditions are reported (including suspicion):
   - Broken or otherwise damaged devices
   - Missing components, device/product parts, single units, or imprints of variable data
   - Deviations in product appearance (e.g., color, odor, spots)

- Contamination or foreign matter present in the package or device
- Deviations in product packaging (e.g., leaflet, primary package (blister), secondary package (folding box))
- Adverse events/incidents (PTC+AE) – in accordance with established Adverse Event principles
- Lack of drug effect (LODE) (PTC+AE) – in accordance with established Lack of Drug Effect principles

Note! Report all IUS breakages as PTC and initiate RQU investigation even if there is no batch number nor sample available.

4. **Reporting of Breakages:**

   All IUS breakages must be reported as PTCs, and an RQU investigation must be initiated, even in the absence of a batch number or sample.

4.2 Instructions for Complaint Sample Shipping

1. **Sample Packaging:**
   - All samples must be properly packaged and labeled in accordance with safety and regulatory requirements.
   - Samples must **not** be cleaned, bent, or otherwise altered during the packaging process.
   - If available, return the original device packaging along with the samples.
2. **Packaging for Biological Substances**
   - For Category B biological substances (e.g., blood-contaminated medical devices), samples must be packaged appropriately, utilizing a double plastic bag and a rigid outer package.
   - All packages must be clearly marked as "Used Medical Device" for sample shipment.
3. **Shipping Documentation**
   - The shipment must include an invoice that details the contents of the shipment.
   - Multiple samples may be combined into a single shipment; however, each sample must be clearly labeled with an identification number linked to the corresponding complaint ID.
4. **Procedure for Absence of Complaint Sample**
   - If a complaint sample is not available for return, select the appropriate rationale for the absence of the sample in the complaint database.
   - Any additional information received after the case entry has been closed may be added to the communication protocol within the complaint database.

Note! Shipping Address for Complaint Samples:

Bayer Oy
Ex Vivo Laboratory/ Complaint samples
Pansiontie 47
FI-20210 Turku
Finland
Tel. +358 2078521

4.3 Error Codes for PTC / PTC+AE / Adverse Event Types for AEs

The initial error code is entered by RQU.
The following table provides guidance for classification of defects related to IUS products. The table contains the most frequently reported product technical complaints related to IUS products.

PTC - IUS Products						
PRODUCT COMPLAINT REASON	**COMPLAINT TYPE**	**OBJECT TYPE**	**ERROR CODE**	**ERROR SUB-CODE**	**RETURN SAMPLE**	**INITIAL RISK CATEGORY***
Tips of T-body arms would not retract into tube	COMPLAINT	IUS	Problem loading	T-body arms would not retract into tube	YES	Low
Releasing problem related to suspected product defect	COMPLAINT	IUS	Problem releasing	IUS did not release from insertion tube	YES	Low
T-body (Other than T-body loop) breakage	COMPLAINT	IUS	Damage to T-body – other than T-body loop	During loading, during insertion, during use / in situ, other	YES	Medium/Low*
T-body (Other than T-body loop) breakage during removal procedure	COMPLAINT	IUS	Damage to T-body – other than T-body loop	During removal	YES	Medium
T-body arms fail to open in situ	COMPLAINT	IUS	Damage to T-body – other than T-body loop	Arms bent	YES	Low
Thread breakage or detachment during removal procedure	COMPLAINT	IUS	Problem with threads	Threads broke/detached during removal	YES	Medium
Thread breakage or detachment	COMPLAINT	IUS	Problem with threads	Threads broke/detached during loading, Threads broke/detached during insertion, Threads broke/detached during use / in situ	YES	Medium/Low*
Issue with threads (no breakage reported)	COMPLAINT	IUS	Problem with threads	other, threads entangled	YES	Low
T-body loop breakage	COMPLAINT	IUS	Damage to T-body loop	During loading, During insertion, During use / in situ, other	YES	Medium/Low*
T-Body loop breakage during removal procedure	COMPLAINT	IUS	Damage to T-body loop	During removal	YES	Medium
Hormone cylinder damaged/displaced/detached	COMPLAINT	IUS	Hormone Cylinder Damaged/Displaced/ Detached	During loading, During insertion, During removal, During use / in situ, other	YES	Medium/Low*
Insertion tube detached / bent	COMPLAINT	IUS	Damage to inserter	Tube bent / detached	YES	Low
Inserter handle damaged / broken	COMPLAINT	IUS	Damage to inserter	Damage to inserter handle, other	YES	Low
Deviating slider movement	COMPLAINT	IUS	Problem with slider	other	YES	Low
Loose / missing flange	COMPLAINT	IUS	Problem with Flange	other	YES	Medium/Low*
Received Damaged - Product (includes all inserter and IUS components) e.g. IUS prematurely released inside primary package	COMPLAINT	IUS	Received Damaged	IUS/Inserter	YES****	Low
Received Damaged – external packaging (Not attributed to shipping)	COMPLAINT	Folding box	Damage	Breakage, Open, Originality seal damaged, Originality seal missing, Other	YES / PICTURES	Low
Primary package received damaged, Lack of sterility (not attributed to shipping)	COMPLAINT	IUS	Received damaged	Sterile packaging, seam open, Sterile packaging, other damages	YES	Medium/Low*
Contamination inside the package or in product e.g. particle inside primary package.	COMPLAINT	IUS	Contamination / Foreign matter	Contamination or foreign material in package / device	YES	Low

(not used for insertion)						
Contamination inside the package or in product e.g. particle inside primary package. (used for insertion)	COMPLAINT	IUS	Contamination / Foreign matter	Contamination or foreign material in package / device	YES	Medium
Missing insertion instructions / leaflet	COMPLAINT	Folding box	Quantity	Leaflet missing	NO	Low
Missing product / component	COMPLAINT	IUS	Quantity	Component missing. Device missing in sterile packaging	YES	Low
Missing lot number– blister	COMPLAINT	IUS	Print/Labelling	Batch missing on sterile packaging	YES / PICTURES	Low
Missing lot number / Expiration date – folding box	COMPLAINT	Folding box	Print/Labelling	Variable data missing	YES / PICTURES	Low
Diverted product	COMPLAINT	Article	Diversion	Smuggling	YES	Medium/Low
Suspected counterfeit	COMPLAINT	Article	Falsification	Imitation, Manipulation	YES	Medium/Low
PRODUCT COMPLAINT REASON	**COMPLAINT TYPE**	**OBJECT TYPE**	**ERROR CODE**	**ADVERSE EVENT TYPE**	**RETURN SAMPLE**	**INITIAL RISK CATEGORY**
LODE (Lack of drug effect) **	COMPLAINT	--	--	Heavy menstrual bleeding (For Mirena only), pregnancy, other	YES	Low
Adverse Event (Incident) related to suspected product defect ***	COMPLAINT	--	--	Expulsion, dislocation, spasm (vaginal, cervical), removal, perforation, procedural pain and bleeding, other	YES	Medium/Low*

* Risk classification based on pharmacovigilance instructions

** Decision based on specific case; see instructions chapter 6.

*** Decision based on specific case; see pharmacovigilance instructions when the case needs to be entered as PTC case.
The instructions for IUS products AE case handling are given in User Letter no 40.

**** For complaints concerning the IUS premature released inside the primary package, if the serial number is available and folding box cannot be sent for investigation, please provide a picture of the serial number on the secondary package. The serial number is needed for tracking purposes.

5 IUS cases reporting only an AE

IUS cases reporting an AE only (without reference to product breakage or deformation, device malfunction or usability issue) are listed in User Letter 40, table 1. They require no complaint database entry nor technical investigation.

6 Handling of IUS LODE cases

Enter all IUS LODE cases as PTC + AE in complaint database if:

- sample provided with or without batch number
- Confirmation of Quality request

No complaint database entry if: No sample provided (with or without batch number).

7 IUS - Usability Issues

Usability Issues (UIs) include, e.g.:

- Product difficult to use (opening of packages, device preparation, device insertion etc.)
- Instructions for use were not followed
- Instructions for use not easy to follow
- Contamination of product prior to use due to incorrect handling
- Deviation from storage conditions
- Device package opened for insertion but was not used (CSI)

In usability issue cases no sample is required. However, if the sample is received, it can be sent for investigation. Usability issues are entered into complaint database, but do not require a technical investigation. If needed, also usability issue cases can be sent for investigation.

In usability issue case entry, select complaint type as UE (Use Error) or CSI (Customer Satisfaction Issue). UE is used when the case is related to incorrect handling of the product. In CSI cases no product malfunction or handling error is reported.

7.1 UI + AE Cases

Usability issue cases associated to a reported AE must be processed as PTC + AE as individual cases. However, there are exceptions, and they are listed in User Letter 40, table 2. These exception cases will be periodically reviewed and monitored by the responsible Quality Assurance functions within Product Supply.

7.2 UI - Error Codes

The following table of usability issues contains examples of most reported usability related issues. Select as specific error code as possible, when reporting usability issues. Each UI entry contains case description.

UI - IUS Products					
COMPLAINT REASON	**ORIGIN**	**COMPLAINT TYPE**	**OBJECT TYPE**	**ERROR CODE**	**ERROR SUB-CODE**
Contaminated prior insertion	Dropped on floor; Patient changed mind after unit opened	CSI	IUS	Customer satisfaction issue	Contaminated prior use by user
Anatomical condition	Non-product related anatomical issue e.g. retroverted uterus; vaginal stenosis, cervical stenosis, unable to sound patient [1]	CSI /UE	IUS	Customer satisfaction issue	Anatomical reasons
Difficulty opening package	Difficulties opening package(s)	CSI	IUS	Customer satisfaction issue	Opening of the package difficult
IUS expelled in HCPs office	Expelled in the office after insertion before the customer has left the office	CSI	IUS	Customer satisfaction issue	Expelled in the office
Difficulty cutting threads	Incorrect handling	UE	IUS	Use error	Difficulty cutting threads
Incorrect insertion technique	Failure to follow Instructions	UE	IUS	Use error	Instructions not followed, Incorrect insertion technique
IUS did not release from inserter	Handling Error; failed insertion and/or complication of insertion	UE	IUS	Use error	Failed insertion, Incorrect insertion technique
IUS was released from inserter, expelled with scissors or speculum.	Incorrect handling	UE	IUS	Use error	IUS came out with scissors or speculum
IUS released prematurely	Failure to follow instructions	UE	IUS	Use error	Instructions not followed,
Threads did not release from inserter or threads entangled	Handling error	UE	IUS	Use error	Threads entangled
Tube bent during loading / insertion	Anatomical issue / Excessive bending of the tube	UE	IUS	Use error	Insertion tube bent

[1] If a temporary stenosis occurred, e.g. during replacement of an IUS or if the insertion is successful at a second attempt, this is considered a medical event and are processed as UE+AE (processed as PTC+AE).

8 History of changes

Version	Significant changes with respect to previous version	Effective date
1	Initial version	2018-07-15
2	Update of error codes for complaints, Added a table for AE specific case entry, Added information that each product related complaint needs to be entered as individual complaint, Added that when no sample is available it needs to mark why it is not available, Instructions for sample shipment has been added, Instructions for Usability case entries has been more specified	2019-12-15
3	Chapter 1 - Scope: Added LODE case handling Chapter 3 - IUS Product Description: Removed specific product indications and time of use (duplicate), text clarification on IUS and EvoInserter. Chapter 3.2 - Insertion Process: Updated according to effective insertion instructions Chapter 4 - IUS - Product Technical Complaints: Text clarification Chapter 4.1 - Instructions for Complaint Sample Shipping: New sub chapter, text moved from chapter 4. Added shipping address. Chapter 4.2 - Error Codes for PTC / PTC+AE: New sub chapter, text moved from chapter 4, Table for error codes and initial risk category updated according to supplement 1582, User Letter 40 and Dev@Com error code update. Table "AE Case Examples from Dev@Com User Letter no 40" removed and replaced with reference to User Letter 40. Chapter 5 - IUS Cases Reporting only an AE: Added new chapter and reference to User Letter 40. Chapter 6 - Handling of LODE Cases: New chapter, clarification on IUS LODE case handling. Chapter 7 - IUS - Usability Issues: Text clarification, removed medication errors from UI as per User Letter 40. Chapter 7.1 - UI + AE Cases: New chapter, added reference to User Letter 40 Chapter 7.2 - UI - Error Codes: New chapter, text moved from chapter 7, updated UI table according to User Letter 40 and Dev@Com error code update. Chapter "Investigation Category Definitions" removed.	2021-04-01
4	Dev@Com replaced with complaint database. IUS Product Description: Silver ring added to the IUS product description. Table providing object and error (sub)codes and guidance of classification for IUS PTC related defects updated: • Added: Complaint reason - Issue with threads (no breakage reported), • Updated: Complaint reason - T-Body loop breakage during removal procedure, Received Damaged – external packaging (Not attributed to shipping), Missing insertion instructions / leaflet, Missing product / component, Missing lot number /	2024-11-01

	Expiration date – blister, Missing lot number / Expiration date – folding box. Handling of LODE cases: LODE cases are entered as PTC + AE in complaint database if sample provided with or without batch number or if confirmation of quality request. Table providing object and error (sub)codes and guidance of classification for IUS usability related issues updated: • Footnote added to the complaint reason - anatomical condition. • Removed: Complaint reason - Product replacement request.	
5	• Transfer to new IQMS template and editorial/formatting changes • The 'Threads broke/detached: other" Error Sub-code was removed from the complaint reason "Thread breakage or detachment". As it is not available any more in the complaint database. Instead, the Complaint reason 'Issue with threads (no breakage reported) " and the Error Sub-code 'Other' can be used. • Added the instruction for providing the serial number from the secondary packages regarding complaints related to the IUS prematurely released inside the primary package. • For Complaint reasons: 'LODE (Lack of drug effect)' and 'Adverse Event (Incident) related to suspected product defect' o The text in the columns 'Object type' and 'Error code' removed. In IQMS there are no error codes for AE-cases. o The column 'Error sub-code' replaced with 'Adverse event type' which is according to IQMS.	See coversheet

Task: Content Approval Verdict: Approve	Mirja Kiiskinen, (mirja.kiiskinen@bayer.com) Approver - Content 10-Jul-2025 10:19:05 GMT+0000

Task: Final Approval Verdict: Approve	Katrin Schnittger, (katrin.schnittger@bayer.com) Approver 10-Jul-2025 10:40:15 GMT+0000